



Mediacrest Training College

Nairobi, Kenya

Advanced Data Analytics & Machine Learning

8-Week Advanced Training Program

Comprehensive Session Notes

Week 4 · Day 1

*Advanced Logistic Regression & Classification Theory:
Statistical Modeling vs. Machine Learning for Heart Disease Prediction*

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“Model complexity should not be treated as a proxy for predictive superiority. For high-stakes clinical applications, a well-performing and interpretable regularized logistic regression model provides the optimal balance between predictive accuracy, clinical trust, accountability, and policy usability.”

About This Document

These comprehensive notes accompany **Week 4, Day 1** of the **Advanced Data Analytics & Machine Learning** program at Mediacrest Training College. This session represents a critical pivot from foundational data preparation to advanced statistical modeling, regularization theory, and machine learning classification.

We will explore the nuanced trade-offs between statistical interpretability (via `statsmodels`) and raw predictive performance (via `scikit-learn`). Key topics include rigorous handling of missing clinical data, avoiding target leakage, formulating parsimonious logistic regression models with interaction terms, and deeply understanding the mathematical foundations of L1 (Lasso) and L2 (Ridge) regularization. The session culminates in a comprehensive policy brief evaluating model fairness, interpretability, and deployment strategies for the Kenyan healthcare system.

Course at a Glance

	8 Weeks (20+ intensive sessions)
Duration	
Core Tools	Python (pandas, NumPy, statsmodels, scikit-learn, matplotlib, seaborn)
Focus	Regularization theory, model comparison, stratified fairness, health policy integration
Capstone	Production-ready ML system with diagnostics, explainability & policy recommendations
Context	Healthcare, Clinical Risk Prediction, and Epidemiological Modeling

Where This Session Fits

Week 4 is titled “**Advanced Logistic Regression & Classification Theory.**”

- Day 1 (this document):** Statistical baseline, transition to ML, deep mathematical dive into L1/L2 regularization, classifier comparison, and comprehensive policy briefing.
- Day 2:** Advanced ensemble methods (Random Forests, Gradient Boosting), hyperparameter tuning, and advanced model interpretability (SHAP values, partial dependence plots).
- Day 3:** Time-to-event analysis (Survival Analysis, Cox Proportional Hazards) and advanced clinical decision curve analysis.

How to Use These Notes

- Follow along in code.** Every listing is runnable. Open the companion Jupyter Notebook and execute each cell as you read.
- Boxes matter.** **Key Concept** boxes hold the theory; **Warning** boxes highlight common pitfalls; **Kenyan Clinical Context** boxes connect material to local healthcare realities.

- **Master the mathematics.** Chapter 4 contains rigorous derivations of regularization. Understanding these is crucial for technical interviews and advanced model debugging.

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1 Data Preparation & Clinical Realities

1.1 Target Audience, Dataset & Dependencies

This module is designed for data analysts and data scientists transitioning to rigorous statistical modeling and clinical decision support. We will use the `Heart.csv` dataset, containing pre-diagnostic clinical cardiovascular measurements for 303 patients.

The required Python dependencies are: `pandas`, `numpy`, `statsmodels`, `scikit-learn`, `matplotlib`, and `seaborn`.

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5 import statsmodels.api as sm
6 import statsmodels.formula.api as smf
7 from sklearn.model_selection import train_test_split, StratifiedKFold,
8     cross_val_score
9 from sklearn.preprocessing import StandardScaler
10 from sklearn.linear_model import LogisticRegression
11 from sklearn.tree import DecisionTreeClassifier
12 from sklearn.metrics import (
13     confusion_matrix, classification_report, roc_auc_score,
14     f1_score, roc_curve, auc
15 )
16 import warnings
17 warnings.filterwarnings("ignore")
18 # Set professional plotting style
19 sns.set_theme(style="whitegrid", context="notebook")
20 plt.rcParams['figure.figsize'] = (10, 6)
21
22 # 1. Load Data, explicitly handling 'NA', '?', 'NaN', and '' as missing values
23 df = pd.read_csv('Heart.csv', na_values=['NA', '?', 'NaN', ''])
```

Listing 1: Loading the Needed Libraries and Setting Professional Style

1.2 Handling Missingness: Clinical Reality

Understanding Missing Data Mechanisms

In clinical datasets, missing data is rarely random. Understanding the mechanism is critical for choosing the right imputation strategy:

- **MCAR (Missing Completely at Random):** Missingness is unrelated to any observed or unobserved data. (Rare in clinical settings).
- **MAR (Missing at Random):** Missingness is related to observed data. For example, older patients might systematically skip the **Ca** (major vessels) test due to mobility issues, but the missingness is explained by their observed **Age**.
- **MNAR (Missing Not at Random):** Missingness is related to the unobserved value itself. For example, patients with severe, symptomatic disease might avoid the **Thal**

(thalassemia) scan due to fear of the procedure, meaning the missingness is directly tied to the severity of the condition.

Clinical Reality

Dropping rows with NA in clinical data (listwise deletion) can introduce severe selection bias. Sicker patients might skip certain tests, meaning dropping them would artificially make the remaining dataset look healthier than the true population. We must impute cautiously to preserve valuable "missingness" information.

1.3 Imputation Strategy & Feature Engineering

An initial audit reveals missing values in **Ca** (ordinal, 0-3) and **Thal** (nominal).

- **For Ca (Ordinal 0-3):** We use median imputation. The median is robust to outliers and preserves the ordinal nature of the major vessel count without artificially inflating or deflating the risk score.
- **For Thal (Nominal):** We create an explicit **unknown** category. This is a powerful technique that preserves the information that the test was missing, allowing the model to learn if "missingness" itself is a predictor of the outcome, thereby avoiding bias.

```
1 # Strategic Imputation
2 df['Ca'] = pd.to_numeric(df['Ca'], errors='coerce')
3 df['Ca'] = df['Ca'].fillna(df['Ca'].median())
4 df['Thal'] = df['Thal'].fillna('unknown')
5
6 # Feature Engineering: Age Group for later stratified fairness analysis
7 df['Age_Group'] = np.where(df['Age'] > 60, '>60', '<=60')
8
9 # Feature Engineering: Sex Labels for intuitive clinical reporting
10 df['Sex_Label'] = df['Sex'].map({0: 'Female', 1: 'Male'})
11
12 print(f"Dataset shape after imputation: {df.shape}")
13 print(f"Missing values remaining: {df.isnull().sum().sum()}")
```

Listing 2: Imputation and Feature Engineering

Pro Tip

Avoid mean imputation for ordinal clinical scores. The mean of an ordinal variable (like a pain scale or vessel count) may result in a non-integer value that has no clinical meaning. Median imputation or treating it as a categorical variable is statistically sounder.

1.4 Target Leakage Prevention

Target Leakage

Ensure no post-diagnosis variables (e.g., results of an angiogram or cardiac catheterization) are used to predict the initial diagnosis. In this dataset, all features are pre-diagnostic clinical measurements (e.g., resting blood pressure, cholesterol, maximum heart rate), making it safe

for predictive modeling.

2 The Statistical Baseline (statsmodels)

2.1 Formulating Logistic Regression with Interaction Terms

We begin by establishing a statistical baseline using `statsmodels`. This library is designed for statistical inference, providing rich diagnostics like p-values, confidence intervals, and odds ratios.

The Dummy Variable Trap

In `statsmodels`, we MUST use `drop_first=True` (or Treatment coding, as shown below) to avoid perfect multicollinearity (the dummy variable trap) when an intercept is present. If we include a dummy variable for every category, their sum equals the intercept column, making the $X^T X$ matrix non-invertible.

```
1 # Treatment coding automatically drops the reference category ('typical')
2 formula = "HD ~ Age + Sex_Label + Age:Sex_Label + ExAng + Oldpeak + C(ChestPain,
3     Treatment('typical'))"
4 # Fit the model using Maximum Likelihood Estimation (MLE)
5 stat_model = smf.logit(formula, data=df).fit(dis=0)
6 print(stat_model.summary())
```

Listing 3: Model Specification with Interaction

2.2 Interpreting the Model Summary

The MLE output provides several critical global metrics:

- **No. Observations:** 303.
- **Pseudo R-squared (McFadden):** 0.3874. While lower than OLS R^2 , a McFadden R^2 between 0.2 and 0.4 is considered an excellent fit for logistic regression, indicating substantial improvement over an intercept-only (null) model.
- **LLR p-value:** 6.289×10^{-31} . This highly significant p-value indicates the model fits the data significantly better than the null model.

2.3 Extracting Odds Ratios and p-values

Coefficients in logistic regression represent the change in the *log-odds* of the outcome. To make them clinically interpretable, we exponentiate them to obtain Odds Ratios (OR).

```
1 params = stat_model.params
2 pvalues = stat_model.pvalues
3 odds_ratios = np.exp(params)
4 conf_int = np.exp(stat_model.conf_int())
5
6 or_df = pd.DataFrame({
7     'Odds Ratio': odds_ratios,
8     'CI Lower': conf_int[0],
9     'CI Upper': conf_int[1],
10    'P-value': pvalues
11 }).round(3)
```

Listing 4: Extracting Adjusted Odds Ratios

Key Clinical Findings:

- **ExAng (Exercise-induced Angina):** $OR = 2.80$. Patients with exercise-induced angina have approximately 2.8 times the odds of heart disease, holding other variables constant.
- **Oldpeak (ST depression):** $OR = 2.14$. A one-unit increase in ST depression is associated with 2.14 times the odds of heart disease.
- **ChestPain (asymptomatic):** $OR = 9.90$. Asymptomatic chest pain is a massive risk multiplier compared to typical angina.

2.4 Checking Statistical Assumptions

Assumption 1: Linearity of the Logit

Logistic regression assumes a linear relationship between continuous predictors and the *log-odds* of the outcome. We verify this by binning **Age** into quartiles, calculating the mean log-odds for each bin, and plotting against the mean age. A roughly straight line confirms the assumption.

Assumption 2: Residual Analysis

We plot Pearson residuals against fitted probabilities. In binary logistic regression, we expect a characteristic "double-band" structure: the upper curve represents patients with heart disease ($HD=1$), and the lower curve represents those without ($HD=0$). Residuals should converge toward zero at the extremes (high confidence predictions), with maximum variance near the center (log-odds ≈ 0 , probability $\approx 50\%$), reflecting maximum uncertainty.

3 Transitioning to Machine Learning (scikit-learn)

3.1 statsmodels (Inference) vs scikit-learn (Prediction)

Paradigm Shift

- **statsmodels:** Optimized for *inference*. Provides rich statistical diagnostics (p-values, CI, ORs) but lacks built-in ML workflows (scaling, cross-validation, regularization).
- **scikit-learn:** Optimized for *prediction* and generalization to unseen data. Requires all categories to be represented (no `drop_first`) and handles the intercept internally.

3.2 Preparing Data for scikit-learn

```
1 # 1. Encoding: Keep ALL dummy columns (drop_first=False)
2 features_ml = ['Age', 'Sex', 'RestBP', 'Chol', 'MaxHR', 'ExAng', 'Oldpeak', 'Ca',
3               'ChestPain', 'Thal']
4 X = pd.get_dummies(df[features_ml], drop_first=False, dtype=int)
5 y = df['HD']
6
7 # 2. Stratified Train/Test Split (maintains class balance)
8 X_train, X_test, y_train, y_test = train_test_split(
9     X, y, test_size=0.25, random_state=42, stratify=y
10 )
```

Listing 5: Encoding and Splitting for ML

3.3 Feature Scaling (Crucial for Regularization)

Why Scaling Matters

Regularization penalties (L1 and L2) are distance-based. If `Chol` is measured in hundreds (e.g., 250) and `Ca` is measured in single digits (e.g., 2), the penalty will disproportionately shrink the coefficient of `Chol` simply because of its scale, not its predictive importance. We must standardize continuous variables to have a mean of 0 and a standard deviation of 1.

```
1 numeric_cols = ['Age', 'RestBP', 'Chol', 'MaxHR', 'Oldpeak', 'Ca']
2 scaler = StandardScaler()
3
4 # Fit on TRAIN data only to prevent data leakage
5 X_train_scaled = X_train.copy()
6 X_test_scaled = X_test.copy()
7 X_train_scaled[numeric_cols] = scaler.fit_transform(X_train[numeric_cols])
8 X_test_scaled[numeric_cols] = scaler.transform(X_test[numeric_cols])
```

Listing 6: Standard Scaling

4 The Mathematics of Regularization (L1 vs L2)

4.1 1. Why Regularization is Needed

Consider the ordinary least squares (OLS) regression model:

$$y = X\beta + \varepsilon$$

where y is the response vector, X is the $n \times p$ predictor matrix, β is the coefficient vector, and ε is the error vector. OLS estimates coefficients by minimizing the residual sum of squares:

$$\hat{\beta}_{OLS} = \arg \min_{\beta} \|y - X\beta\|_2^2$$

The closed-form solution is:

$$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y$$

The Problem: When predictors are highly correlated (multicollinearity) or when p is large relative to n , $(X^T X)$ becomes nearly singular. OLS coefficients can become unstable, excessively large, and highly sensitive to small changes in the data. Regularization addresses this by adding a penalty for large coefficients.

4.2 2. General Regularized Regression Framework

A general penalized regression objective can be written as:

$$\min_{\beta} \left[\underbrace{\|y - X\beta\|_2^2}_{\text{Model fit}} + \lambda \underbrace{P(\beta)}_{\text{Penalty}} \right]$$

where $\lambda \geq 0$ is the regularization parameter controlling the strength of the penalty. The difference between Ridge and Lasso lies entirely in the choice of $P(\beta)$.

4.3 3 & 4. L2 Regularization (Ridge) and its Derivation

Ridge regression uses the squared (L_2)-norm penalty: $P(\beta) = \sum_{j=1}^p \beta_j^2$. Ignoring the intercept for simplicity, the loss function is:

$$L(\beta) = (y - X\beta)^T (y - X\beta) + \lambda \beta^T \beta$$

Expanding the first term:

$$L(\beta) = y^T y - 2\beta^T X^T y + \beta^T X^T X \beta + \lambda \beta^T \beta$$

Differentiating with respect to β and setting to zero:

$$\frac{\partial L}{\partial \beta} = -2X^T y + 2X^T X \beta + 2\lambda \beta = 0$$

Dividing by 2 and rearranging yields the fundamental Ridge estimator:

$$\hat{\beta}_{Ridge} = (X^T X + \lambda I)^{-1} X^T y$$

The addition of λI to the diagonal of $X^T X$ guarantees the matrix is invertible, solving the multicollinearity problem.

4.4 5. Why Ridge Shrinks Coefficients

Suppose predictors are standardized and mutually orthogonal, so $X^T X = I$. Then:

$$\hat{\beta}_{Ridge} = (I + \lambda I)^{-1} X^T y = \frac{1}{1 + \lambda} X^T y$$

Since $\hat{\beta}_{OLS} = X^T y$, we obtain:

$$\hat{\beta}_{Ridge,j} = \frac{1}{1 + \lambda} \hat{\beta}_{OLS,j}$$

Every coefficient is multiplied by the same shrinkage factor $\frac{1}{1+\lambda}$. For example, if $\lambda = 3$, a coefficient of 8 becomes $8 \times 0.25 = 2$. *Important qualification:* Ridge continuously shrinks coefficients toward zero but generally **does not** set them exactly equal to zero.

4.5 6 & 7. L1 Regularization (Lasso) and Exact Zeros

Lasso uses the absolute value (L_1)-norm: $P(\beta) = \sum_{j=1}^p |\beta_j|$. For a single standardized predictor where $\sum x_i^2 = 1$ and $z = X^T y$, the Lasso objective is:

$$L(\beta) = \frac{1}{2}(z - \beta)^2 + \lambda|\beta|$$

For $\beta > 0$, $|\beta| = \beta$. Differentiating and setting to zero gives $\beta = z - \lambda$ (valid only if $z > \lambda$).

For $\beta < 0$, $|\beta| = -\beta$. Differentiating and setting to zero gives $\beta = z + \lambda$ (valid only if $z < -\lambda$).

4.6 8. The Soft-Thresholding Solution

Combining the cases yields the soft-thresholding operator $S(z, \lambda)$:

$$\hat{\beta}_{Lasso} = \begin{cases} z - \lambda, & z > \lambda \\ 0, & |z| \leq \lambda \\ z + \lambda, & z < -\lambda \end{cases} \quad \text{or equivalently} \quad \hat{\beta}_{Lasso} = \text{sign}(z)(|z| - \lambda)_+$$

where $(a)_+ = \max(a, 0)$. This equation mathematically proves why Lasso performs feature selection: if the unregularized signal z is smaller than the penalty λ , the coefficient is driven to **exactly zero**.

4.7 11. Geometric Explanation

Regularization can be viewed as constrained optimization: $\min_{\beta} \|y - X\beta\|^2$ subject to a constraint on β .

- **Ridge (L_2):** Constraint $\sum \beta_j^2 \leq t$ creates a circular/spherical region. The optimum (where

the error ellipse touches the circle) is smooth and unlikely to land exactly on an axis.

- **Lasso (L_1):** Constraint $\sum |\beta_j| \leq t$ creates a diamond-shaped region (in 2D). The sharp corners lie exactly on the coordinate axes. The error ellipse frequently touches the diamond at a corner, resulting in $\beta_j = 0$.

4.8 12. Derivative Explanation (Subgradient)

For L_2 , $P'(\beta) = 2\beta$, so $P'(0) = 0$. The penalty's gradient vanishes at zero.

For L_1 , the ordinary derivative at zero does not exist. Instead, the subgradient is $\partial|\beta|_{\beta=0} = [-1, 1]$. This creates a *range* of values that can balance the gradient of the loss function, making $\beta = 0$ a valid, optimal solution. This is the deeper mathematical reason behind Lasso's sparsity.

```
1 # L2 (Ridge) - shrinks coefficients evenly
2 lr_l2 = LogisticRegression(penalty='l2', C=1.0, solver='lbfgs', max_iter=1000,
   random_state=42)
3
4 # L1 (Lasso) - shrinks and performs feature selection
5 lr_l1 = LogisticRegression(penalty='l1', C=1.0, solver='saga', max_iter=1000,
   random_state=42)
6
7 lr_l2.fit(X_train_scaled, y_train)
8 lr_l1.fit(X_train_scaled, y_train)
```

Listing 7: L1 vs L2 Implementation in scikit-learn

5 Comparing Classifiers & Evaluation

5.1 Defining Models & Cross-Validation

We compare a linear, regularized model against a non-linear tree-based model using 5-Fold Stratified Cross-Validation. Stratification ensures the class ratio (Heart Disease prevalence) is maintained in each fold, providing a robust estimate of generalization performance.

```

1 models = {
2     "Logistic Regression (L2)": LogisticRegression(penalty='l2', C=1.0, solver='
    lbfgs', max_iter=1000, random_state=42),
3     "Decision Tree (Non-linear)": DecisionTreeClassifier(max_depth=4,
    min_samples_split=10, random_state=42)
4 }
5
6 cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
7
8 for name, model in models.items():
9     # Scale data for LR, but NOT for Decision Tree (trees are scale-invariant)
10    X_cv = X_train_scaled if "Logistic" in name else X_train
11    roc_scores = cross_val_score(model, X_cv, y_train, cv=cv, scoring='roc_auc')
12    f1_scores = cross_val_score(model, X_cv, y_train, cv=cv, scoring='f1')
13    print(f"{name}: Mean ROC-AUC: {roc_scores.mean():.3f} (+/- {roc_scores.std()
    :.3f})")

```

Listing 8: Stratified K-Fold Cross-Validation

Results:

- **Logistic Regression (L2):** Mean ROC-AUC = 0.895 ($\pm$ 0.054), Mean F1 = 0.799 ($\pm$ 0.051).
- **Decision Tree:** Mean ROC-AUC = 0.841 ($\pm$ 0.075), Mean F1 = 0.766 ($\pm$ 0.079).

Observation: Logistic Regression provides stable, well-calibrated probabilities. The Decision Tree captures non-linear thresholds but is more prone to variance (higher standard deviation) and ultimately underperforms in this dataset.

5.2 Independent Test Set Evaluation

Model	ROC-AUC	F1-Score	Confusion Matrix
Logistic Regression (L2)	0.934	0.849	$\begin{bmatrix} 34 & 7 \\ 4 & 31 \end{bmatrix}$
Decision Tree (Non-linear)	0.811	0.704	$\begin{bmatrix} 30 & 11 \\ 10 & 25 \end{bmatrix}$

Table 1: Test Set Performance Comparison

Clinical Relevance of Metrics

In heart disease screening at a facility like Kenyatta National Hospital, a high Precision-Recall balance is crucial. A False Negative (missed diagnosis) could result in an unmanaged myocardial infarction. A False Positive (unnecessary angiogram) costs approximately 20,000 Ksh and carries procedural risk. The Logistic Regression model's lower False Negative count (4 vs 10) makes it clinically superior.

6 Interpretability vs. Predictive Power

6.1 Explainability Comparison

- **Logistic Regression:** Provides global, directional, magnitude-based explanations via Odds Ratios. Top features by absolute coefficient: `Sex` (1.090), `ChestPain_asymptomatic` (1.046), `Ca` (1.041).
- **Decision Tree:** Provides local, non-linear, hierarchy-based explanations via Gini Importance. Top features: `Thal_normal` (0.414), `ChestPain_asymptomatic` (0.200).

6.2 Stratified Fairness Check

A model might perform well overall but fail for specific demographic subgroups (algorithmic bias). We evaluate the best performing model (Logistic Regression) on different subgroups of the TEST set.

Subgroup	N	ROC-AUC
Overall	76	0.934
Male	50	0.902
Female	26	0.980
Age ≤ 60	58	0.950
Age > 60	18	0.901

Table 2: Stratified Model Performance (ROC-AUC by Subgroup)

Fairness Directive

The model maintained an ROC-AUC above 0.90 in every reported subgroup. However, variation between groups (e.g., Female 0.980 vs Male 0.902) should not be interpreted as evidence of fairness by itself. Subgroup-specific discrimination is only one component of model equity. Future validation must additionally examine sensitivity, specificity, positive/negative predictive values, and calibration across these groups before deployment in the Kenyan population.

7 Comprehensive Policy Brief

POLICY BRIEF: Statistical Interpretability vs. Machine Learning Performance

To: Ministry of Health (MOH), Kenya – Non-Communicable Diseases (NCD) Directorate

From: Advanced Data Analytics Team, Mediacrest Training College

Date: 2nd September 2026

Subject: Comparative Evaluation of Logistic Regression and ML Classifiers for Heart Disease Risk Stratification

1. Executive Summary

This brief evaluates whether complex machine learning models provide sufficient predictive gains to justify sacrificing the interpretability of logistic regression. Findings indicate that regularized logistic regression provides the optimal balance. Five-fold stratified CV produced a mean ROC-AUC of 0.895 for L2-regularized logistic regression, compared with 0.841 for a decision tree. On the independent test set, the selected model achieved an overall ROC-AUC of 0.934.

2. Policy Context

Deployment of predictive models in healthcare presents a fundamental challenge: Should the health system prioritize maximum predictive performance, or a model whose decisions can be clearly understood and clinically justified? In high-stakes medical applications, a model that produces a slightly higher predictive score but cannot adequately explain its predictions may be less suitable than a statistically interpretable model with comparable discrimination.

3. Key Statistical Findings

The statistical model identified exercise-induced angina (ExAng, OR = 2.80) and ST depression (Oldpeak, OR = 2.14) as important predictors. An Age × Sex interaction suggested that the association between age and heart disease differs somewhat by sex, necessitating sex-specific risk assessment protocols.

4. Interpretability–Performance Trade-Off

Logistic regression offers transparent coefficients, clinically interpretable odds ratios, and straightforward statistical inference. Tree-based approaches offer automatic interaction discovery but lack inherent transparency. In this evaluation, the decision tree achieved a lower cross-validated ROC-AUC (0.841) than regularized logistic regression (0.895). Increased model complexity did not translate into improved performance.

5. Recommended Deployment Strategy

Primary model: Regularized Logistic Regression. It provides strong discrimination, transparent risk-factor relationships, interpretable odds ratios, and greater clinical/regulatory transparency.

Secondary models: Retain complex ML models for benchmarking and prospective validation. A complex model should replace the logistic model only if it demonstrates a clinically meaningful and reproducible improvement.

6. Recommended Validation Before Deployment

Before operational use, the model must undergo:

1. **External validation** on an independent Kenyan clinical dataset.
2. **Calibration assessment** (Calibration slope/intercept) to ensure predicted probabilities match observed prevalence.
3. **Subgroup validation** monitoring performance by sex, age group, and geographic setting.
4. **Clinical utility assessment** via decision-curve analysis to determine net benefit across threshold probabilities.

7. Governance and Monitoring

The model should not be treated as an autonomous diagnostic system. The recommended operational framework is: **Model prediction** → **Clinical review** → **Clinical decision**. A monitoring framework must periodically assess discrimination drift, calibration drift, and changes in predictor distributions.

8. Policy Recommendation

Based on current evidence, regularized logistic regression is recommended as the primary candidate model.

The recommendation is based on the combination of: **Strong discrimination + Interpretability + Parsimony + Clinical transparency**. Model complexity should not be treated as a proxy for predictive superiority.

8 Best Practices, Checklist & Deliverables

8.1 Clinical Decision Support Report Checklist

The final output of this pipeline must be a structured, reproducible report for stakeholders. Ensure your final deliverable includes:

- **Data Preparation:** Explicit documentation of missingness handling (e.g., `Ca` median, `Thal` unknown) and encoding strategies.
- **Model Fit & Odds Ratios:** Reporting of Pseudo R-squared and top predictors with their ORs and 95% Confidence Intervals.
- **Diagnostics:** Confirmation of multicollinearity checks (Max VIF < 5), linearity of logit, and interaction term significance.
- **Evaluation Metrics:** ROC-AUC, Precision-Recall AUC, and recommended policy thresholds (e.g., 0.481 for balanced, 0.658 for high precision specialist referral).
- **Prevalence Trap Warning:** Explicit statement that ORs overstate Risk Ratios (RR) when prevalence is high (~50%), and that marginal predicted probabilities should be used for patient-facing communication.

8.2 Next Steps & Recommendations

Actionable Directives for Health Systems

1. **HMIS Integration:** Integrate this parsimonious model into the national Health Management Information System (HMIS) or develop a lightweight mobile tool for Community Health Promoters (CHPs) for initial triage.
2. **Prospective Pilot:** Conduct a pilot study in 2-3 sub-county hospitals to validate these scenario-specific thresholds on local Kenyan demographic data before nationwide rollout.
3. **Bias Monitoring:** Ensure continuous monitoring for algorithmic bias across gender and age groups. Disparate impact must be actively measured and mitigated.
4. **Model Retraining:** Establish a pipeline for periodic model retraining as new clinical data becomes available, ensuring the model adapts to shifting population health trends.

Final Deliverable: The Policy Memo

Draft a 1-page technical memo addressed to the County Director of Health. Your memo must include:

1. **Model Justification:** Briefly explain why a parsimonious regularized logistic regres-

sion was chosen over a black-box decision tree, and how missingness was handled to preserve clinical integrity.

2. **Diagnostic Validation:** Report the key statistical findings (ORs for ExAng and Old-peak) and explicitly state whether the model is stable across demographic groups based on the stratified fairness check.
3. **Policy Recommendation:** Recommend specific classification thresholds for (a) Community Health Promoter screening (maximize recall) and (b) County Referral Hospital triage (maximize precision), justifying the trade-offs in terms of cost (e.g., 20,000 Ksh angiogram) and patient safety.

9 Glossary of Key Terms

Health Management Information System (HMIS) A systematic framework for collecting, processing, and using health data to support decision-making at all levels of the health system.

Maximum Likelihood Estimation (MLE) The method used to estimate the parameters of a logistic regression model by maximizing the likelihood of observing the given data.

Odds Ratio (OR) The ratio of the odds of an event occurring in one group to the odds of it occurring in another group. Prone to exaggeration when outcome prevalence is high.

Parsimonious Model A statistical model that achieves a good fit with the fewest possible predictor variables, reducing the risk of overfitting and improving interpretability.

Precision-Recall AUC The area under the Precision-Recall curve, highly informative for evaluating models where false positives/negatives have asymmetric costs (common in clinical screening).

Risk Ratio (RR) The ratio of the probability of an outcome in an exposed group to the probability in an unexposed group. Preferred over OR for clinical communication when prevalence is high.

Soft-Thresholding The mathematical operator in Lasso regression that shrinks coefficients toward zero and sets them to exactly zero if the unregularized signal is smaller than the penalty λ .

Stratified Cross-Validation A variation of k-fold cross-validation that returns stratified folds: each set contains approximately the same percentage of samples of each target class as the complete set.

10 References & Further Reading

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